

Technical Report: System Requirements & Infrastructure Specifications

Project: India Air Quality & Climate Digital Twin

Document Series: Capstone Report Materials — Part 3

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1. Hardware Requirements

The system has been engineered to run efficiently on standard developer workstations as well as enterprise cloud virtual machines:

Parameter	Minimum (Inference & Dashboard)	Recommended (Training & Ingestion)	High-Resolution Enterprise Scaling (0.05°)
CPU Architecture	x86_64 or ARM64 (Apple Silicon / Graviton)	x86_64 (Intel Core i7/i9 or AMD Ryzen 7/9)	16+ Core Intel Xeon / AMD EPYC
CPU Cores	4 Physical Cores	8 Physical Cores	16 to 32 Cores
System Memory (RAM)	8 GB RAM	16 GB to 32 GB RAM	64 GB to 128 GB RAM
Storage Capacity	10 GB Available SSD Space	50 GB to 100 GB NVMe SSD	500 GB+ High IOPS NVMe SSD
GPU Acceleration	Not required (CPU Fallback implemented)	NVIDIA GPU (RTX 3060/4060, 8 GB VRAM)	NVIDIA A100 / H100 (40/80 GB VRAM)
Network Bandwidth	10 Mbps (for interactive tiles)	50+ Mbps (satellite bulk downloads)	1 Gbps Dedicated Cloud Pipe

2. Software & Operating System Specifications

2.1 Supported Operating Systems

- Microsoft Windows:** Windows 10 / Windows 11 (64-bit), PowerShell 7+ or Windows Terminal.
- Linux:** Ubuntu 20.04 / 22.04 LTS, Debian 11+, RHEL / Rocky Linux 9.
- macOS:** macOS Monterey (12.0) or higher (Native Apple Silicon & Intel support).

2.2 Python Runtime Environment

- Python Version:** Python 3.10 to 3.13 (Tested and verified on Python 3.13.0 64-bit).
- Virtual Environment Tool:** venv or conda / mamba.

2.3 Core Library Stack & Versions

- **Numerical & Tabular:**

- numpy >= 1.26.0
- pandas >= 2.2.0
- scipy >= 1.12.0
- pyarrow >= 14.0.0

- **Machine Learning & Deep Learning:**

- scikit-learn >= 1.4.0
- xgboost >= 2.0.0
- lightgbm >= 4.3.0
- torch >= 2.1.0 (PyTorch CPU / CUDA)
- joblib >= 1.3.0

- **Geospatial & Remote Sensing:**

- shapely >= 2.0.0
- scipy.spatial/scipy.interpolate

- **Web & API Framework:**

- fastapi >= 0.110.0
- uvicorn >= 0.28.0
- pydantic >= 2.6.0
- requests >= 2.31.0
- python-dotenv >= 1.0.0

- **Interactive Visualization:**

- streamlit >= 1.32.0
- pydeck >= 0.8.0
- plotly >= 5.19.0

- **Testing & Quality Assurance:**

- pytest >= 8.0.0

3. External API Credentials & Accounts (Optional / Live Mode)

The digital twin includes complete offline and simulated operational fallbacks. To run live data synchronization from orbital satellites, the following free research accounts are supported:

1. **Copernicus Data Space Ecosystem (CDSE):**

- Access to Sentinel-5P TROPOMI Level 2 & Level 3 products.
- Credentials configured in .env: CDSE_USERNAME, CDSE_PASSWORD.

2. ECMWF Climate Data Store (CDS):

- Access to ERA5 hourly and daily reanalysis products.
- Credentials configured in .env: CDSAPI_URL, CDSAPI_KEY.

3. NASA Earthdata Login:

- Access to MODIS MAIAC AOD (MCD19A2), LST (MOD11A1), and MERRA-2.
- Credentials configured in .env: EARTHDATA_USERNAME, EARTHDATA_PASSWORD.

4. NASA FIRMS (Fire Information for Resource Management System):

- Access to near-real-time VIIRS active fire detections.
- Credentials configured in .env: FIRMS_MAP_KEY.